Junk on the Moon and Mars

An inventory of every human-made object on two worlds,
built from Wikipedia's lists: 133 objects on the Moon and 27 on Mars

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Abstract

I build a single, mapped inventory of the human-made objects on the Moon and Mars from English Wikipedia's lists of artificial objects on the two bodies, its list of landings on extraterrestrial bodies, and the articles that name the small items left by the Apollo crews. The inventory holds 133 objects on the Moon (104 table rows and 29 small items) and 27 on Mars (20 rows and 7 smaller pieces). Counting each listed mass once, 203,228 kg of hardware lies on the Moon, 58% of it American and 31% Soviet or Russian, against 10,380 kg on Mars. Five spent Saturn V third stages, crashed on purpose, make up 34% of the lunar total. Arrivals came in two waves: 34 objects in the 1960s and 28 in the 1970s, none between 1977 and 1989, and 20 already in the 2020s, from 7 countries and several private companies. Of 40 attempted soft landings on the Moon, 28 (70%) succeeded; on Mars, 11 of 16 (69%), 9 of 10 for the United States. Only 15 of the 87 located lunar objects lie on the far side, and only three lunar and two Martian objects are listed as still operating, although six retroreflectors on the Moon are still used for laser ranging. I document how the tables were parsed, where the sources disagree, and what the inventory leaves out.

Keywords: Moon; Mars; space exploration; spacecraft; lunar landings; Mars landings; space heritage; Wikipedia; open data; mapping

1 Introduction

Since the Soviet probe Luna 2 struck the Moon on 13 September 1959, spacecraft from a growing number of countries, and lately from private companies, have landed on, crashed into or been abandoned on the Moon and Mars. The hardware they left is an unusual archive: a record of engineering success and failure, of national programmes and their priorities, and of objects that some have argued deserve protection as heritage. Yet there is no single authoritative public inventory of it. Space agencies document their own missions; the most convenient public compilations are the lists maintained by Wikipedia's editors.

This paper turns those lists into one consistent, mapped table and asks simple questions of it. How many objects are on each body, and how much do they weigh? Who left them, and when? How many arrived intact, and how many still work? How often do soft landings succeed? Where are they, and how many lie on the Moon's far side? The paper accompanies an interactive map (https://maps.tristanhotham.com/moon-mars/).

Table 1: Objects and mass by country or agency. Mass in kilograms as listed, each figure counted once; small items have no mass.

	Moon			Mars		
Who	Objects	kg	Share	Objects	kg	Share
United States	80	117,861	58.0%	16	5,189	50.0%
Soviet Union and Russia	23	63,535	31.3%	5	3,055	29.4%
China	12	15,052	7.4%	3	1,526	14.7%
Japan	10	3,046	1.5%			
India	3	3,258	1.6%			
Europe (ESA)	1	307	0.2%	3	610	5.9%
Others (Israel, Luxembourg, UAE)	4	169	0.1%			
All	133	203,228	100%	27	10,380	100%

2 Data

Object tables. The article "List of artificial objects on the Moon" (Wikipedia, 2026a) contains a table of 104 objects with country or agency, year, mass in pounds and kilograms, status, location and notes; its rows are coloured by a legend (crashed unintentionally; impactor or post-mission crash; landed; operational), and a final row gives the article's total. "List of artificial objects on Mars" (Wikipedia, 2026b) contains 20 rows with year, agency, mission, objects, mass, status and location, coloured success, operational or failure.

Landings. "List of landings on extraterrestrial bodies" (Wikipedia, 2026c) lists, for the Moon and Mars, each landing or impact with its date, coordinates and notes, coloured by outcome. I use it for arrival dates and for the success of soft landing attempts.

Small items. The Moon list states that its table omits "lesser" Apollo objects such as tools, retroreflectors, the science packages and the commemorative and personal objects left by the astronauts. I add 29 such items named in the introduction of that list and in the articles "Moon landing" (Wikipedia, 2026d), "Fallen Astronaut" (Wikipedia, 2026e), "Apollo Lunar Surface Experiments Package" (Wikipedia, 2026f) and "Lunar Roving Vehicle" (Wikipedia, 2026g), and 7 pieces named in the text of the Mars list. "List of missions to the Moon" and "List of missions to Mars" (Wikipedia, 2026h,i) were consulted for context. Three titles requested for this study (on golf on the Moon, the Apollo flags and lunar laser ranging) do not exist under those names and are not used.

Base maps. Static figures and the interactive map use the OpenPlanetaryMap lunar and Martian basemaps (OpenPlanetary, 2026), built from NASA and USGS data.

3 Methods

3.1 Parsing

Tables were read from the parsed HTML with BeautifulSoup and expanded cell by cell, honouring row and column spans, so that a value written once for two rows is attached to both. Footnote markers were removed. Positions come from the decimal coordinates embedded in each location cell. The Mars table gives some longitudes as west longitudes beyond 180 degrees (Mars 2 at 313°W) or as east longitudes from 0 to 360 (Opportunity at 354.47°E); all longitudes were wrapped to the range -180 to 180. Objects without coordinates (17 on the Moon) keep none.

3.2 Mass

Masses are the tables' kilograms. Where a mass cell spans two rows (Hakuto-R Mission 1 and the Emirates Lunar Mission it carried) it is counted once, and where a rover row repeats its lander's figure exactly (Chang'e 3 and Yutu; Chang'e 4 and Yutu-2) the figure is counted once. Upper and lower bounds (Luna 20 and 24 descent stages; the Chang'e 5 and 6 ascent stages) and masses the table marks as launch masses (Luna 8) or as measured before a capsule was released (Luna 9) are taken as given. Small items have no mass. The sum, 203,228 kg, differs from the article's own table total (222,729 kg) and from the figure in its introduction ("over 187,400 kilograms"); I report all three.

3.3 Type and status

Each row was classified by its name and status as a rocket or ascent stage (S-IVB third stages, the Luna 2 and Chang'e 5-T1 rocket stages, the LCROSS Centaur, a Falcon 9 upper stage, the Apollo Lunar Module and Chang'e ascent stages, and stages that crashed after separation), a rover (including small hoppers and Mars's Ingenuity helicopter), an impact or crash (a hard arrival that is not a stage) or a lander (a soft arrival). Rows that combine a lander and a rover, such as "Luna 17 and Lunokhod 1", are kept as one object, as in the source. Status follows the row colours. On the Moon, soft-landed means coloured landed or operational. On Mars, it also includes Mars 3 and Beagle 2, which the table colours as failures but whose status text says they reached the surface intact. "Still working" means coloured operational.

3.4 Landing success

A soft landing attempt is a row of the landings list that the article colours as a soft landing with data return, sample return or crewed landing (a success), or whose note describes an unsuccessful or failed landing, lost contact or a crash, excluding deliberate impacts. Every Mars row is an attempt. Landers that never reached their target, such as those lost at launch, are not in that list and are not counted. Countries are grouped as in Table 1; Europe on Mars combines Beagle 2 (United Kingdom with ESA) and Schiaparelli (ESA with Roscosmos).

3.5 Small items and positions

Each added item is supported by a sentence in its source article, which the build checks. Items are placed at the coordinates of their landing site (the Apollo descent stage, or the Mars lander), except the Fallen Astronaut and its plaque, which the article locates itself. Their positions therefore show the site, not the exact spot; some, such as heat shields and the Curiosity counterweights, lie kilometres away. Far side means a longitude more than 90 degrees east or west, the definition used in the Moon list.

4 Results

4.1 How many, and how heavy

The inventory holds 133 objects on the Moon and 27 on Mars (Figures 1 and 2). The Moon's 104 table rows comprise 28 landers, 10 rovers, 47 impacts or crashes and 19 stages; the table names

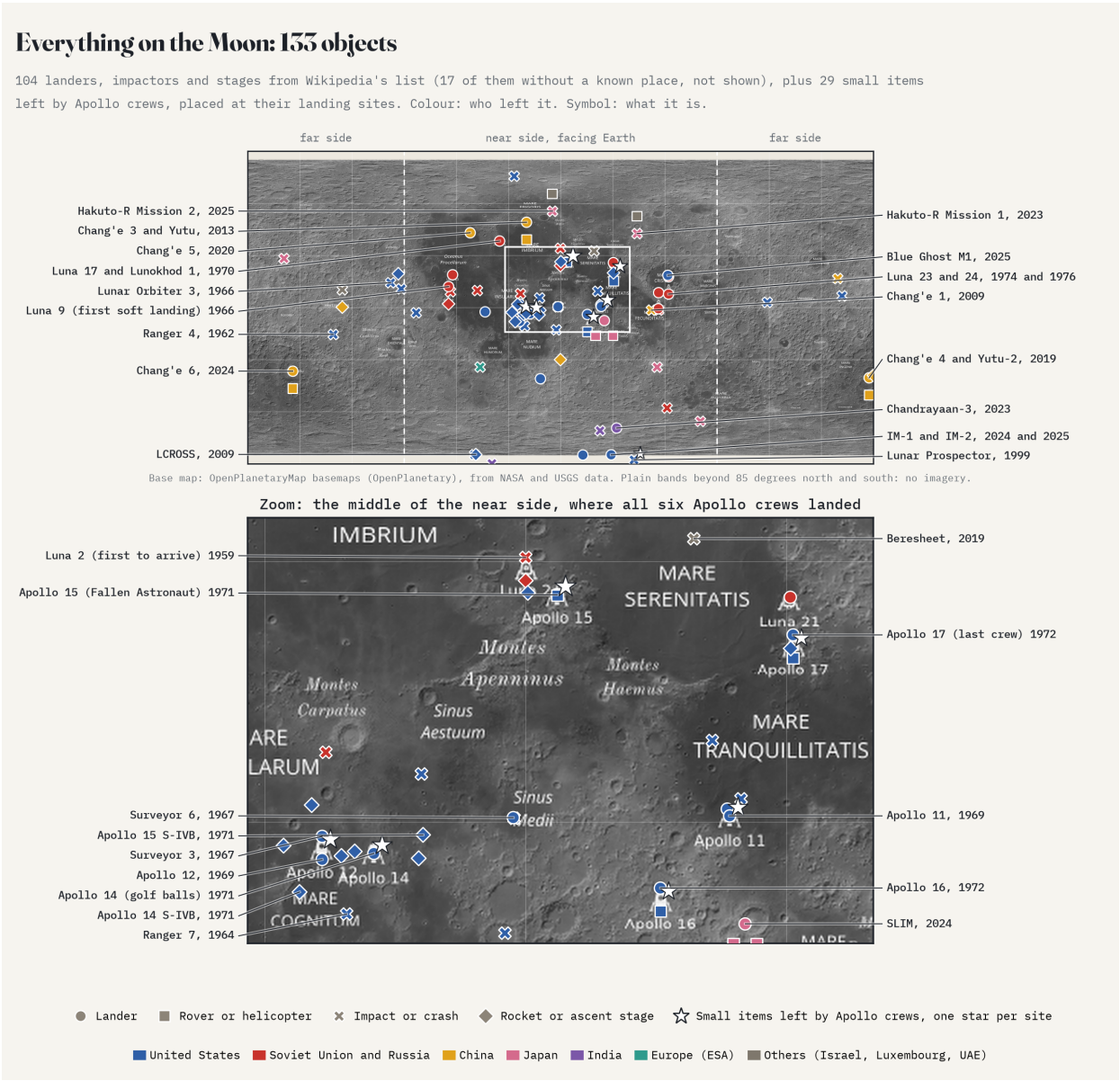


Figure 1: Every located object on the Moon, by type and by who left it, with a zoom on the centre of the near side. Small items are drawn as one star per site. Base map: OpenPlanetaryMap.

17 rovers, hoppers and similar vehicles in all, counting those inside combined rows. Mars has 8 landers, 7 rovers and a helicopter, and 4 crash sites. Objects from 10 countries or agencies have reached the Moon.

Counting each mass once, 203,228 kg lies on the Moon and 10,380 kg on Mars (Table 1, Figure 3). The United States left 58.0% of the lunar total, the Soviet Union and Russia 31.3% and China 7.4%. Apollo hardware accounts for 49.2%, and the five S-IVB stages of Apollo 13 to 17, crashed for seismic research (Wikipedia, 2026c), for 69,468 kg (34.2%). They are the five heaviest objects on either body (Table 2, Figure 4), followed by the 9,100 kg rocket stage that accompanied Luna 2. On Mars the heaviest object is the Tianwen-1 lander (1,285 kg) and the United States left 50.0% of the total.

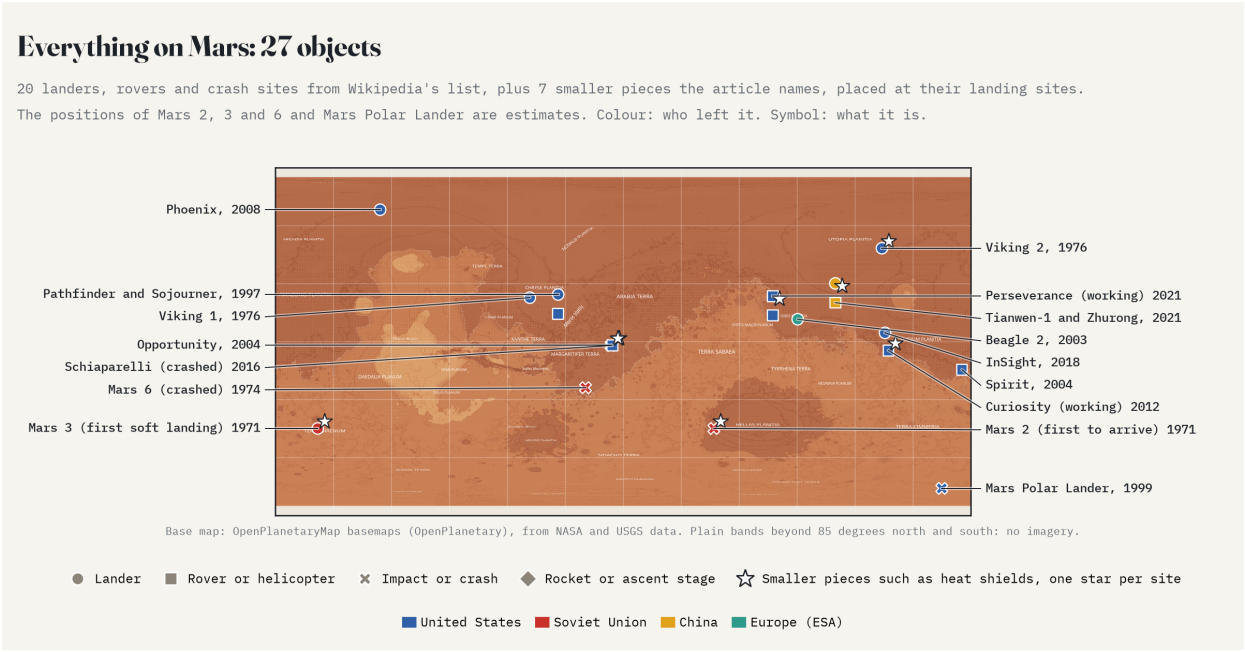


Figure 2: Every object on Mars. The positions of Mars 2, 3 and 6 and Mars Polar Lander are estimates. Base map: OpenPlanetaryMap.

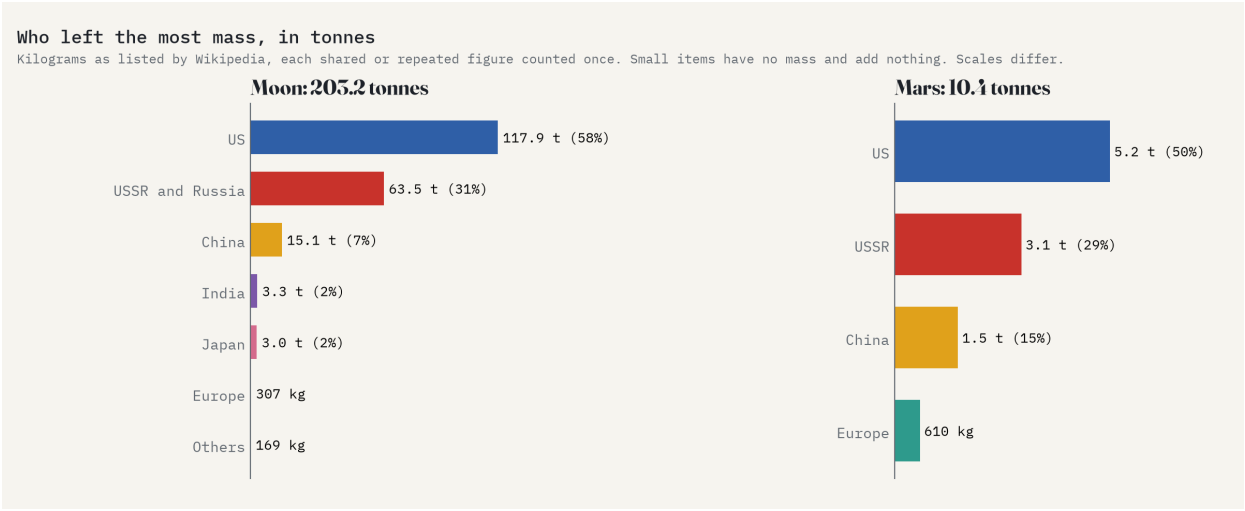


Figure 3: Mass left on the Moon and Mars by country or agency, each listed figure counted once.

Table 2: The ten heaviest objects, kilograms as listed.

Object	Body	Year	kg
Apollo 15 S-IVB (S-IVB-510)	Moon	1971	14,036
Apollo 14 S-IVB (S-IVB-509)	Moon	1971	14,016
Apollo 16 S-IVB (S-IVB-511)	Moon	1972	14,002
Apollo 17 S-IVB (S-IVB-512)	Moon	1972	13,960
Apollo 13 S-IVB (S-IVB-508)	Moon	1970	13,454
Luna 2 rocket third stage (Vostok)	Moon	1959	9,100
Luna 24 descent stage	Moon	1976	5,800
Luna 20 descent stage	Moon	1972	5,727
Luna 17 and Lunokhod 1	Moon	1970	5,600
Luna 23	Moon	1974	5,600

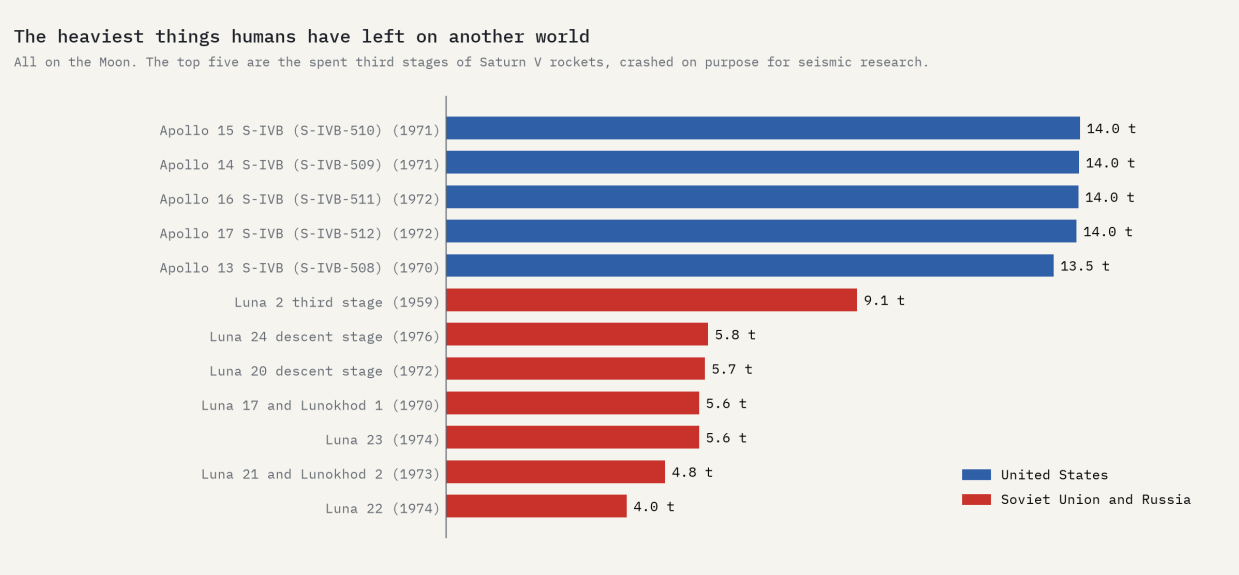


Figure 4: The twelve heaviest objects, all on the Moon.

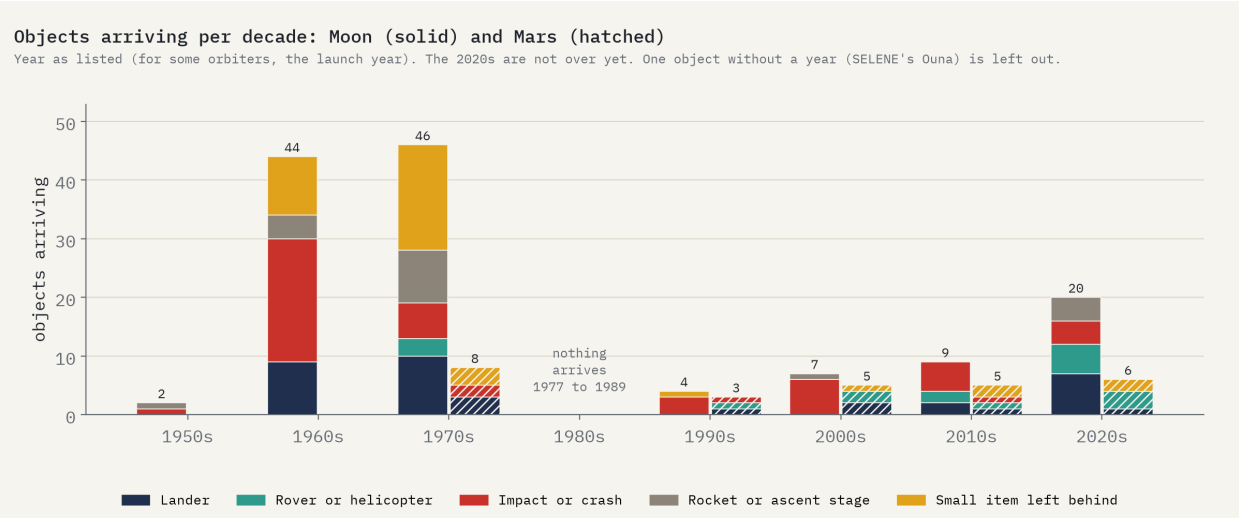


Figure 5: Objects arriving on the Moon and Mars per decade, by type. Years as listed in the tables.

4.2 When

Arrivals came in two waves (Figure 5). The first object, Luna 2, struck the Moon on 13 September 1959; 34 table objects arrived in the 1960s and 28 in the 1970s. After Luna 24 in 1976 no listed object reached the Moon until 1990. The 2000s brought 7 and the 2010s 9, mostly hard impacts; the 2020s have already brought 20. Of these 9 objects in the table were built or flown by companies (SpaceIL, LuxSpace, ispace and ispace Europe, Intuitive Machines, Firefly Aerospace and SpaceX), and 3 of them landed softly. The first object on Mars arrived on 27 November 1971 (Mars 2); Mars received 5 table objects in the 1970s and 5 so far in the 2020s.

4.3 Crashed or landed, and still working

Of the 104 Moon rows, 36 (35%) arrived softly and 68 hard; the article's own summary row speaks of "40 major objects landed softly", a broader count that the row colours do not reproduce. Of the 20 Mars rows, 16 arrived softly (Figure 6). Only three lunar objects are coloured operational, all

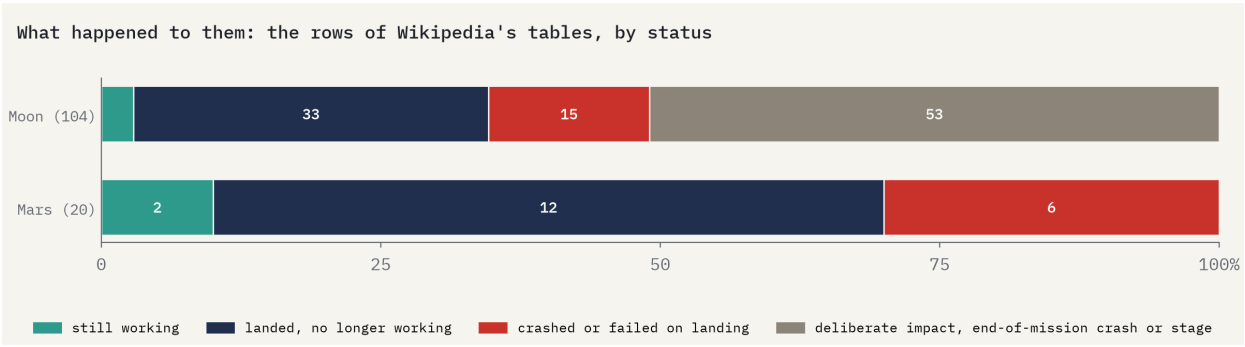


Figure 6: Status of every row of the two tables, from the row colours.

Table 3: Soft landing attempts and successes by country (List of landings on extraterrestrial bodies).

Body	Who	Attempts	Succeeded	Rate	First attempt
Moon	United States	16	14	88%	1966
Moon	Soviet Union and Russia	14	8	57%	1965
Moon	China	4	4	100%	2013
Moon	Japan	3	1	33%	2023
Moon	India	2	1	50%	2019
Moon	Israel	1	0	0%	2019
Moon	All	40	28	70%	
Mars	United States	10	9	90%	1976
Mars	Soviet Union	3	1	33%	1971
Mars	Europe (ESA)	2	0	0%	2003
Mars	China	1	1	100%	2021
Mars	All	16	11	69%	

Chinese (the Chang'e 3 and Chang'e 4 landers and the Yutu-2 rover); the article's introduction also names SLIM as still in use. On Mars, Curiosity and Perseverance are operating. The Moon list adds that the retroreflectors left by Apollo 11, 14 and 15 and carried by Lunokhod 1, Lunokhod 2 and Chandrayaan-3 are still used for laser ranging, and the ALSEP article that they are the only Apollo experiments still in use; the five ALSEP stations were supported until 30 September 1977.

4.4 Landing success

The landings list records 40 soft landing attempts on the Moon, of which 28 (70.0%) succeeded, and 16 on Mars, of which 11 (68.8%) succeeded (Table 3, Figure 7). The first successful lunar soft landing was Luna 9, 3 February 1966. China succeeded in all 4 of its lunar attempts and in its single Mars attempt; the Soviet Union and Russia in 8 of 14 on the Moon and 1 of 3 on Mars. Since 2019, 8 of 13 lunar attempts have succeeded, and 3 of the 6 commercial ones.

4.5 Near side and far side

Of the 87 lunar table objects with a position, 15 (17.2%) lie on the far side, although it is half of the surface. Most of them hit the surface hard; the only soft arrivals there are China's Chang'e 4 lander and Yutu-2 rover and the Chang'e 6 lander and its Jinchan rover.

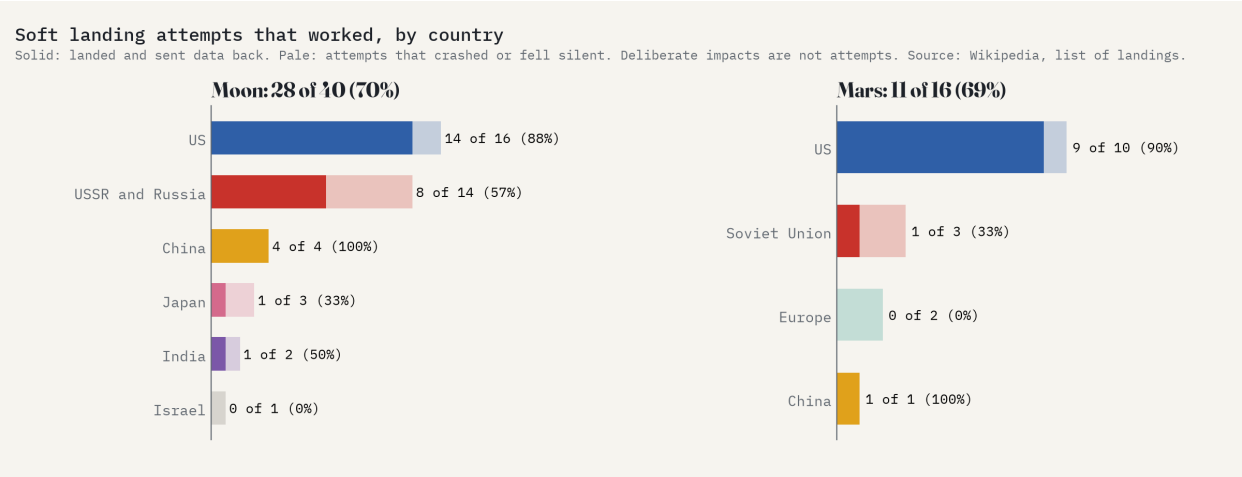


Figure 7: Soft landing attempts and successes by country.

4.6 Small items

The 29 small lunar items (Table 4) include the six American flags, of which "five of the six ... are still standing" according to LROC images (Wikipedia, 2026d); the six ladder plaques; the Apollo 11 goodwill messages disc; Alan Bean's silver astronaut pin; Alan Shepard's golf balls; David Scott's Bible; the 8.9 cm Fallen Astronaut and its plaque naming 8 American astronauts and 6 Soviet cosmonauts (Wikipedia, 2026e); three laser ranging retroreflectors; the EASEP and five ALSEP packages (Wikipedia, 2026f); a discarded fuel transfer tool; and Eugene Shoemaker's ashes, carried by Lunar Prospector. Some objects often mentioned, such as the falcon feather from Apollo 15 or Charles Duke's family photograph, are not named in these sources and are not included.

Table 4: Small items added from the articles' text, placed at their landing sites.

Item	Body	Mission	Year
Apollo 11 United States flag	Moon	Apollo 11	1969
Apollo 12 United States flag	Moon	Apollo 12	1969
Apollo 14 United States flag	Moon	Apollo 14	1971
Apollo 15 United States flag	Moon	Apollo 15	1971
Apollo 16 United States flag	Moon	Apollo 16	1972
Apollo 17 United States flag	Moon	Apollo 17	1972
Apollo 11 commemorative plaque on the Lunar Module ladder	Moon	Apollo 11	1969
Apollo 12 commemorative plaque on the Lunar Module ladder	Moon	Apollo 12	1969
Apollo 14 commemorative plaque on the Lunar Module ladder	Moon	Apollo 14	1971
Apollo 15 commemorative plaque on the Lunar Module ladder	Moon	Apollo 15	1971
Apollo 16 commemorative plaque on the Lunar Module ladder	Moon	Apollo 16	1972
Apollo 17 commemorative plaque on the Lunar Module ladder	Moon	Apollo 17	1972
Apollo 11 goodwill messages disc	Moon	Apollo 11	1969
Silver astronaut pin left by Alan Bean for Clifton C. Williams	Moon	Apollo 12	1969
Golf balls hit by Alan Shepard	Moon	Apollo 14	1971
Bible left by David Scott	Moon	Apollo 15	1971
Fallen Astronaut statuette	Moon	Apollo 15	1971
Fallen Astronaut memorial plaque (14 names)	Moon	Apollo 15	1971
Apollo 11 Laser Ranging Retroreflector	Moon	Apollo 11	1969

Table 5: Objects whose positions in the object tables and in the landings list differ by more than one degree of arc.

Object	Object table	Landings list	Degrees
Luna 5	8.00N 23.00W	31.00S 8.00W	41.6
Luna 8	9.10N 63.30W	9.60N 62.00W	1.4
Lunar Orbiter 1	6.70N 162.00E	6.35N 160.72E	1.3
Surveyor 2	5.50N 12.00W	4.00S 11.00W	9.6
IM-1 Odysseus lander and EagleCam	84.90S 12.90E	80.13S 1.44E	5.0
IM-2 Athena lander, with MAPP LV1, Micro-Nova, AstroAnt and Yaoki	84.79S 29.20E	85.00S 31.00W	5.1
Mars 2 lander and PrOP-M rover	45.00S 47.00E	45.00S 30.00W	52.2
Mars Polar Lander and Deep Space 2 probes	76.00S 165.00E	73.00N 150.00E	149.3

Table 4: Small items added from the articles' text, placed at their landing sites.

Item	Body	Mission	Year
Apollo 14 Laser Ranging Retroreflector	Moon	Apollo 14	1971
Apollo 15 Laser Ranging Retroreflector	Moon	Apollo 15	1971
Apollo 11 Early Apollo Scientific Experiments Package (EASEP)	Moon	Apollo 11	1969
Apollo 12 Apollo Lunar Surface Experiments Package (ALSEP)	Moon	Apollo 12	1969
Apollo 14 Apollo Lunar Surface Experiments Package (ALSEP)	Moon	Apollo 14	1971
Apollo 15 Apollo Lunar Surface Experiments Package (ALSEP)	Moon	Apollo 15	1971
Apollo 16 Apollo Lunar Surface Experiments Package (ALSEP)	Moon	Apollo 16	1972
Apollo 17 Apollo Lunar Surface Experiments Package (ALSEP)	Moon	Apollo 17	1972
Apollo 12 ALSEP fuel transfer tool (discarded)	Moon	Apollo 12	1969
Ashes of Eugene Shoemaker (on Lunar Prospector)	Moon	Lunar Prospector	1999
Soviet pennant on the Mars 2 lander	Mars	Mars 2	1971
Soviet pennant on the Mars 3 lander	Mars	Mars 3	1971
Viking 2 surface sampler shroud	Mars	Viking 2	1976
Opportunity heat shield	Mars	MER-B	2004
Curiosity (MSL) counterweights ejected during descent	Mars	Mars Science Laboratory	2012
Schiaparelli heat shields and parachute	Mars	ExoMars 2016	2016
Three Forks Sample Depot (Perseverance sample tubes)	Mars	Mars 2020	2022

4.7 Where the sources disagree

For 86 objects both the object tables and the landings list give coordinates. They differ by more than one degree of arc for 8 (Table 5), among them Luna 5, Surveyor 2, IM-1 and IM-2, Mars 2 and Mars Polar Lander. Some differences reflect uncertain crash sites; the Mars Polar Lander positions are in different hemispheres. I keep the object tables' positions throughout and publish the alternatives in the data.

5 Limitations

First, Wikipedia is an edited compilation, not a primary record, and it changes: the inventory describes the revisions cited. Second, the tables are partial by their own account; the Mars list omits springs, fragments, parachutes and heat shields, and many small Apollo items (tools, bags, equipment) are not named anywhere in these sources. Third, masses mix dry and launch masses and include bounds, so totals are approximate. Fourth, the type classification is mine and some rows hold several objects. Fifth, small items are placed at their landing sites. Sixth, web maps cannot show latitudes beyond about 85 degrees, so the interactive map draws polar objects at its edge; the static figures and the page's polar chart show them properly. Seventh, landing success counts only attempts that reached their target.

6 Conclusion

About 0 tonnes of hardware lie on the Moon and about 0 on Mars. Most of the lunar mass is the legacy of a few years of the Cold War race, and a third of it is five rocket stages. Almost all of it is dead. The 2020s are adding objects faster than any decade since the 1970s, from more countries and from companies, with mixed results: landing on the Moon is still far from routine, and Mars remains harder. A consistent, open inventory makes those patterns visible, and shows how much of this archive is still described only loosely.

Data and code availability

The interactive map is at https://maps.tristanhotham.com/moon-mars/. Data and code are at https://github.com/Tristan-Hotham/moon-mars and archived with this paper at https://doi.org/10.5281/zenodo.23872995. The object table derived from Wikipedia is released under CC BY-SA 4.0; other derived data and text under CC BY 4.0; code under the MIT licence.

AI statement

I wrote the text, with drafting and editing assistance from Claude (Anthropic), which also helped prepare the analysis code. I reviewed and checked all text and results.

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